

Supplementary Material

Community Knowledge

Table 1. Definitions of factors affecting the Clear Creek and Klaza caribou from community knowledge data entry.

Category	Variable(s)	Definition
Human disturbance	Human disturbance	General "human disturbance", type not defined
Mining and Exploration	quartz mining, placer mining, quartz and placer mining, exploration, helicopter disturbance, trenches	Activities related to placer and quartz mining and exploration.
Helicopter Disturbance	Helicopter disturbance	Previously under mining and exploration (most helicopter disturbance is associated with Quartz exploration), but separating because helicopters were mentioned often.
Road and trails	roads, trails, hunting access, all-season roads, traffic	Impacts related to roads and access issues (typically increased access). Includes references to hunting pressure, which was interpreted as a problem of increased access
Fire	wildfire	References to fire, wildfire
Climate	insects, food availability, winter kill	References to climate related issues
Predation	wolf predation, grizzly bear predation	Predation on the caribou

Competition	Bison, Elk	Competition between caribou and other species; or other ungulate species pushing the caribou out of certain areas
Snow	Snow depth	Snow related variables and issues

Caribou Location Data

Table 2. Number of GPS (Early Winter Late Winter, Summer, Fall Rut seasons) and VHF (Snow, Snow-free seasons) collar locations and individual caribou included in seasonal RSFs for the Clear Creek and Klaza caribou herds in Yukon, Canada.

Herd	Season	Number of individuals	Number of locations
Klaza	Snow	28	130
	Snow-free	28	190
	Early winter	12	893
	Late winter	12	795
	Summer	11	857
	Fall-rut	11	351
Clear Creek	Snow	21	128
	Snow-free	21	128
	Early winter	26	1465
	Late winter	29	1004
	Summer	30	1284
	Fall-rut	27	620

RSF Covariates

Table 3. Breakdown of reclassified landcover

Value	Reclassified Name	Original classes
1	Conifer Forest	Temperate Conifer, Taiga Conifer
5	Deciduous / Mixed wood Forest	Deciduous Forest, Mixedwood Forest
8	Shrublands	Shrublands
10	Grasslands	Grasslands, Shrubland-lichen-moss, Grassland-lichen-moss, Cropland
14	Wetland	Wetland
16	Non-vegetated	Barren, Ice/snow, Urban, Water

Table 4. Description of disturbance types included in each human development covariate and what each covariate represents.

Covariate Type	Description	Represents
Buffered human disturbance	All linear and areal surface disturbance with different buffers to test different zones of influence.	All human surface disturbance, most of it associated with resource extraction (mining). Sensory disturbance increased in snow-free seasons. Linear features may facilitate predator movement.

Buffered mining disturbance	Areal mining related features with different buffers to test different zones of influence. Disturbance types mining, placer mining – minor, placer mining – significant, quartz mining – minor and quartz mining – significant.	Quartz and Placer mining operations. Includes advanced exploration programs. Sensory disturbance increased in snow-free seasons.
Buffered roads and trails	Linear access features with different buffers to test different zones of influence. Disturbance types access road, access assumed, arterial road, local road, and trail.	Roads and trails. Access for miners and hunters. Sensory disturbance increased in snow-free seasons. Linear features may facilitate predator movement.
Disturbance density	Sum of all linear and areal surface disturbances in each grid cell. Different grid sizes were used to test different zones of influence.	The amount or intensity of surface disturbance in a given area. Most surface disturbance is associated with resource extraction (mining). Sensory disturbance increased in snow-free seasons. Linear features may facilitate predator movement, but most predators probably avoid areas with higher

		densities especially in snow-free seasons.
Linear feature density	Density of all linear features in different search radiuses to test different zones of influence.	Density of linear features, most of which are roads and trails (access) and survey lines associated with mining/exploration programs. Linear features may facilitate predator movement, but most predators probably avoid areas with higher LF densities especially in snow-free seasons.

Table 5. Covariates used in seasonal RSFs for the Clear Creek and Klaza caribou in Yukon, Canada. Note that landcover, buffered mining, buffered roads and trails, and disturbance count were not used in the VHF-based models.

Covariate	Scale (m)	Type	Units
Elevation	30	Continuous	Elevation (m)
Slope	30	Continuous	Slope (degrees)
Fire	30	Categorical	Burned/unburned
Lichen cover	30, 100, 250	Categorical	Lichen cover/no lichen cover
Conifer cover	30, 100, 250	Categorical	Conifer cover/no cover
Deciduous shrub cover	30, 100, 250	Categorical	Deciduous shrub cover/no cover
Graminoid cover	30, 100, 250	Categorical	Graminoid cover/no cover
Landcover	30, 100, 250	Categorical	Conifer forest, deciduous/mixed wood forest, shrublands, grasslands, wetlands, non-vegetated
Buffered surface disturbances	30, 250, 500, 1000, 2000, 3000	Categorical	Disturbed/undisturbed
Buffered mining disturbances	30, 250, 500, 1000, 2000, 3000	Categorical	Disturbed/undisturbed
Buffered roads and trails	30, 250, 500, 1000, 2000, 3000	Categorical	Disturbed/undisturbed
Disturbance count	250, 500, 1000, 2000, 3000	Continuous	Number (count) of disturbances

Linear feature density	250, 500, 1000, 2000, 3000	Continuous	Linear feature density (km/km ²)
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AICc Tables

Klaza Caribou Herd

Table 6. Disturbance AICc Model Selection for the Klaza caribou snow season featuring the five best fitting models and the base environmental model.

Model	K	AICc	Delta AICc	AICcWt	Cum.Wt	LL
Glm8e + lineden 2000	7	840.72	0.00	0.22	0.22	-413.31
Glm8e + lineden 3000	7	841.52	0.79	0.15	0.37	-413.70
Glm8e + distbuff 1000 m	7	841.69	0.96	0.14	0.51	-413.79
Glm8e + distbuff 30 m	7	841.94	1.22	0.12	0.63	-413.92
Glm8e + lineden 1000	7	842.05	1.32	0.11	0.74	-413.97
Glm8e	6	844.49	3.77	0.03	0.94	-416.20

Table 7. Disturbance Anick model selection for Klaza caribou snow-free season featuring the five best fitting models and the base environmental model.

Model	K	AICc	Delta AICc	AICcWt	Cum.Wt	LL
Glm9e + lineden 1000	7	1152.3	0.00	0.22	0.22	-569.12
	2					
Glm9e + lineden 2000	7	1152.4	0.15	0.21	0.43	-569.20
	7					

Glm9e + lineden 3000	7	1153.3 5	1.03	0.13	0.57	-569.64
Glm9e + lineden 1000 + dist buff 30	8	1153.5 9	1.27	0.11	0.61	-568.74
Glm9e + dist buff 30	7	1153.8 9	1.57	0.10	0.67	-569.90
Glm9e	6	1154.4 1	1.74	0.09	0.75	-571.00

Table 8. Disturbance AICc model selection for Klaza caribou early winter season featuring the five best fitting models and the base environmental model.

Model	K	AICc	Delta AICc	AICcWt	Cum.Wt	LL
Glm1e + mine 3000 m buff + road 500 m buff	14	1776.77	0.00	0.55	0.55	-874.34
Glm1e + mine 3000 m buff + road 500 m buff + distden 2000	15	1778.78	2.01	0.20	0.76	874.34
Glm1e + mine 3000 m buff	13	1779.25	2.48	0.16	0.92	876.58
Glm1e + mine 3000 m buff + distden 2000	14	1780.79	4.02	0.07	0.99	876.35
Glm1e + mine 2000 m buff	13	1785.38	8.61	0.01	1.00	879.65
Glm1e	12	1807.42	30.65	0.00	1.00	-891.67

Table 9. Disturbance AICc model selection for Klaza caribou late winter season featuring the five best fitting models and the base environmental model.

Model	K	AICc	Delta AICc	AICcWt	Cum.Wt	LL
Glm1e + lineden 3000 + dist 3000 m buff	1 6	4354.5 4	0.00	0.99	0.99	- 2161.2 4

Glm1e + lineden 3000	1 5	4363.5 0	8.95	0.01	1.00	- 2166.7 2
Glm1e + distden 3000 + dist 3000 m buff	1 6	4365.9 9	11.45	0.00	1.00	- 2166.9 6
Glm1e + distden 3000	1 5	4418.0 6	63.52	0.00	1.00	- 2194.0 0
Glm1e + dist 3000 m buff	1 5	4423.2 5	68.70	0.00	1.00	- 2196.5 9
Glm1e	1 4	4619.2 9	264.75	0.00	1.00	- 2295.6 2

Table 10. Disturbance AICc model selection for Klaza caribou summer season featuring the five best fitting models and the base environmental model.

Model	K	AICc	Delta AICc	AICcWt	Cum.Wt	LL
Glm2e + distden 1000 + mine 500 m buff	15	4720.17	0.00	0.91	0.91	- 2345.07
Glm2e + distden 1000	14	4724.76	4.59	0.09	1.00	- 2348.36
Glm2e + lineden 250	14	4733.42	13.26	0.00	1.00	- 2352.70
Glm2e + lineden 3000	14	4733.46	13.29	0.00	1.00	- 2352.71
Glm2e + mine 500m buff	14	4734.83	14.66	0.00	1.00	- 2353.40
Glm2e	13	4760.20	40.03	0.00	1.00	- 2367.09

Table 11. Disturbance AICc model selection for Klaza caribou fall-rut season featuring the five best fitting models and the base environmental model.

Model	K	AICc	Delta AICc	AICcWt	Cum.Wt	LL
Glm10e + lineden 1000	12	579.27	0.00	0.50	0.50	-277.54
Glm10e + dist 1000m buff	12	580.85	1.58	0.23	0.72	-278.33
Glm10e + road 1000 m buff	12	583.25	3.99	0.07	0.79	-279.53
Glm10e + lineden 500	12	584.24	4.97	0.04	0.83	-280.02
Glm10e + lineden 3000	12	584.32	5.05	0.04	0.87	-280.07

Glm10e	11	592.04	12.77	0.00	1.00	-284.94
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Clear Creek Caribou Herd

Table 12. Disturbance AICc model selection for Clear Creek caribou snow season featuring the five best fitting models and the base environmental model.

Model	K	AICc	Delta AICc	AICcWt	Cum.Wt	LL
Glm 2e + dist buff 3000 m	6	533.27	0.00	0.55	0.55	-260.57
Glm 2e + dist buff 3000 m + lineden 1000	7	535.32	2.04	0.20	0.75	-260.57
Glm 2e + dist buff 2000 m	6	535.65	2.38	0.17	0.92	-261.76
Glm 2e + dist buff 1000 m	6	538.84	5.56	0.03	0.95	-263.35
Glm 2e + dist buff 500 m	6	539.86	6.59	0.02	0.97	-263.87
Glm2e	5	542.89	9.62	0.00	0.99	-266.40

Table 13. Disturbance AICc model selection for Clear Creek caribou snow-free season featuring the five best fitting models and the base environmental model.

Model	K	AICc	Delta AICc	AICcWt	Cum.Wt	LL
Glm3e + lineden 2000	6	517.67	0.00	0.41	0.41	-252.76
Glm3e + distbuff 2000 m	6	518.90	1.24	0.22	0.63	-253.38
Glm3e + lineden 3000	6	519.65	1.98	0.15	0.78	-253.75
Glm3e + lineden 1000	6	521.55	3.88	0.06	0.84	-254.70
Glm3e + distbuff 1000 m	6	522.49	4.82	0.04	0.87	-255.18
Glm3e	5	523.36	5.69	0.02	0.93	-256.63

Table 14. Disturbance AICc model selection for Clear Creek early winter season featuring the five best fitting models and the base environmental model.

Model	K	AICc	Delta AICc	AICcWt	Cum.Wt	LL
Glm2e + dist 1000 m buff * elevation	16	5211.55	0.00	1.00	1.00	- 2589.73
Glm2e + lineden 1000	14	5243.20	31.65	0.00	1.00	- 2607.57
Glm2e + dist 1000 m buff	14	5253.08	41.54	0.00	1.00	- 2612.51
Glm2e + distden 1000	14	5253.26	41.71	0.00	1.00	- 2612.60
Glm2e + lineden 500	14	5253.51	41.96	0.00	1.00	- 2612.72
Glm2e	13	5275.70	64.15	0.00	1.00	- 2624.82

Table 15. Disturbance AICc model selection for Clear Creek late winter season featuring the five best fitting models and the base environmental model.

Model	K	AICc	Delta AICc	AICcWt	Cum.Wt	LL
Glm3e + lineden 3000 * elevation	15	3161.06	0.00	0.94	0.94	- 1565.48
Glm3e + dist 1000m buff * elevation	15	3166.69	5.63	0.06	1.00	- 1568.29

Glm3e + lineden 2000 * elevation	15	3176.18	15.13	0.00	1.00	- 1573.04
Glm3e + dist 2000 m buff * elevation	15	3180.06	19.00	0.00	1.00	- 1574.98
Glm3e + dist 3000 m buff * elevation	15	3180.73	19.67	0.00	1.00	- 1575.31
Glm3e	12	3242.14	81.08	0.00	1.00	- 1609.04

Table 16. Disturbance AICc model selection for Clear Creek caribou summer season featuring the five best fitting models and the base environmental model.

Model	K	AICc	Delta AICc	AICcWt	Cum.Wt	LL
Glm3e + road 1000 m buff * dem	16	2925.57	0.00	0.37	0.37	- 1446.74
Glm3e + lineden 3000 * dem	16	2926.32	0.75	0.26	0.63	- 1447.12
Glm3e + distden 3000	14	2927.44	1.87	0.15	0.77	- 1449.69
Glm3e + dist 1000 m buff * dem	16	2927.86	2.29	0.12	0.89	- 1447.89
Glm3e + lineden 2000 * dem	16	2929.21	3.64	0.06	0.95	- 1448.56
Glm3e	13	2947.03	21.46	0.00	1.00	- 1460.49

Table 17. Disturbance AICc model selection for Clear Creek caribou fall-rut season featuring the five best fitting models and the base environmental model.

	K	AICc	Delta AICc	AICcWt	Cum.Wt	LL
Glm4e + distden 1000 + mine 500m buff	14	1416.44	0.00	0.12	0.12	-694.15
Glm4e + mine 500 m buff	13	1416.66	0.21	0.11	0.23	-695.27
Glm4e + distden 1000	13	1416.68	0.23	0.11	0.33	-695.28
Glm4e + dist 30 m buff	13	1417.87	1.43	0.06	0.39	-695.87
Glm4e + road 30 m buff	13	1417.90	1.45	0.06	0.45	-695.89
Glm4e	12	1419.07	2.63	0.04	0.67	-697.48

RSF Regression Tables

Klaza Caribou Herd

Table 18. Estimated regression parameters, standard errors, z-values and P-values for the 1995 snow (VHF telemetry) binomial GLM for the Klaza caribou herd in Yukon, Canada.

	Estimate	Std. error	z value	P-value
Intercept	-1.954	0.123	-15.852	<0.05
Slope	-0.437	0.105	-4.157	<0.05
Lichen 250 m	0.265	0.144	1.838	0.066
Conifer 30 m	0.415	0.110	3.783	<0.05
Graminoid 250 m	0.360	0.127	2.844	<0.05
Burns ≤50 yrs	-0.721	0.269	-2.682	<0.05
Disturbance 1000 m	-0.207	0.100	-2.067	<0.05

Table 19. Estimated regression parameters, standard errors, z-values and P-values for the 1995 snow-free (VHF telemetry) binomial GLM for the Klaza caribou herd in Yukon, Canada.

	Estimate	Std. error	z value	P-value
Intercept	-1.963	0.101	-19.455	<0.05
Elevation	2.028	0.847	2.395	<0.05
Elevation^2	-1.111	0.768	-1.446	0.148
Slope	0.283	0.082	3.471	<0.05
DeciduousShrub 30 m	0.217	0.103	2.114	<0.05
Graminoid 30 m	0.261	0.105	2.485	<0.05
Line density 1000 m	-0.196	0.121	-1.624	0.104

Table 20. Estimated regression parameters, standard errors, z-values and P-values for the 2015 early winter binomial GLM for the Klaza caribou herd in Yukon, Canada.

	Estimate	Std. error	z value	P-value
Intercept	-5.597	0.350	-15.987	<0.05
Elevation	15.006	1.632	9.193	<0.05
Elevation^2	-10.959	1.297	-8.452	<0.05
Deciduous/mixed 30 m	-0.938	0.341	-2.753	<0.05
Shrublands 30 m	-0.300	0.201	-1.496	0.135
Grasslands 30 m	0.664	0.184	3.610	<0.05
Wetland 30 m	1.198	0.558	2.147	<0.05
Non-vegetated 30 m	-1.165	0.411	-2.837	<0.05
Lichen 100 m	1.212	0.081	14.872	<0.05
Conifer 250 m	-0.518	0.057	-9.142	<0.05

Graminoid 30 m	0.353	0.082	4.324	<0.05
Burns $\leq$ 50 yrs	-0.606	0.115	-5.269	<0.05
Mining 3000 m	-0.412	0.115	-3.583	<0.05
RoadsTrails 500 m	-0.173	0.084	-2.057	<0.05

Table 21. Estimated regression parameters, standard errors, z-values and P-values for the 2015 late winter binomial GLM for the Klaza caribou herd in Yukon, Canada.

	Estimate	Std. error	z value	P-value
Intercept	-4.561	0.167	-27.298	<0.05
Elevation	17.465	0.964	18.123	<0.05
Elevation^2	-14.776	0.831	-17.785	<0.05
Slope	0.812	0.172	4.720	<0.05
Slope^2	-1.191	0.183	-6.520	<0.05
Deciduous/mixed 30 m	-0.396	0.128	-3.090	<0.05
Shrublands 30 m	0.243	0.108	2.260	<0.05
Grasslands 30 m	0.628	0.100	6.303	<0.05
Wetland 30 m	-1.332	0.333	-4.000	<0.05
Non-vegetated 30 m	-14.638	193.081	-0.076	0.940
Lichen 30 m	0.422	0.049	8.683	<0.05
Conifer 250 m	0.315	0.052	6.042	<0.05
Graminoid 250 m	0.313	0.045	6.995	<0.05
Burns ≤50 yrs	-0.523	0.054	-9.624	<0.05
Line density 3000 m	-1.446	0.248	-5.819	<0.05
Disturbance 3000 m	-0.222	0.067	-3.298	<0.05

Table 22. Estimated regression parameters, standard errors, z-values and P-values for the 2015 summer binomial GLM for the Klaza caribou herd in Yukon, Canada.

	Estimate	Std. error	z value	P-value
Intercept	-5.492	1.533	-3.583	<0.05
Elevation	9.556	0.661	14.456	<0.05
Elevation^2	-6.151	0.530	-11.597	<0.05
Slope	0.681	0.152	4.471	<0.05
Slope^2	-0.589	0.139	-4.249	<0.05
Deciduous/mixed 250 m	0.047	0.180	0.260	0.795
Shrublands 250 m	1.665	0.108	15.390	<0.05
Grasslands 250 m	1.068	0.113	9.476	<0.05
Wetland 250 m	0.032	0.275	0.118	0.906
Non-vegetated 250 m	0.533	0.179	2.977	<0.05
Lichen 250 m	0.890	0.047	19.088	<0.05
Conifer 250 m	-0.528	0.038	-14.059	<0.05
DeciduousShrub 250 m	0.411	0.047	8.849	<0.05
Disturbance density 1000 m	-0.240	0.072	-3.337	<0.05
Mining 500 m	-1.165	16.770	-0.069	0.945

Table 23. Estimated regression parameters, standard errors, z-values and P-values for the 2015 fall rut binomial GLM for the Klaza caribou herd in Yukon, Canada.

	Estimate	Std. error	z value	P-value
Intercept	-7.313	0.656	-11.142	<0.05
Elevation	12.994	2.447	5.311	<0.05
Elevation^2	-8.547	1.901	-4.496	<0.05
Deciduous/mixed 100 m	-0.682	0.827	-0.825	0.410
Shrublands 100 m	2.534	0.335	7.563	<0.05
Grasslands 100 m	1.959	0.317	6.176	<0.05
Wetland 100 m	-15.038	786.039	-0.019	0.985
Non-vegetated 100 m	2.236	0.533	4.194	<0.05
Lichen 100 m	0.978	0.138	7.088	<0.05
Conifer 30 m	-0.381	0.124	-3.086	<0.05
DeciduousShrub 30 m	0.641	0.173	3.709	<0.05
Line density 1000 m	-1.113	0.335	-3.328	<0.05

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Table 24. Estimated regression parameters, standard errors, z-values and P-values for the 2000 snow season (VHF telemetry) binomial GLM for the Clear Creek caribou herd in Yukon, Canada.

	Estimate	Std. error	z value	P-value
Intercept	-1.847	0.131	-14.124	<0.05
Slope	-0.172	0.133	-1.299	0.194
Lichen 250 m	0.345	0.140	2.455	<0.05
Conifer 100 m	0.508	0.164	3.094	<0.05
Graminoid 100 m	-0.010	0.117	-0.083	0.934
Disturbance 3000 m	0.350	0.101	3.456	<0.05

Table 25. Estimated regression parameters, standard errors, z-values and P-values for the 2000 snow-free (VHF telemetry) binomial GLM for the Clear Creek caribou herd in Yukon, Canada.

	Estimate	Std. error	z value	P-value
Intercept	-1.839	0.135	-13.601	<0.05
Slope	0.186	0.121	1.542	0.123
Lichen 30 m	0.330	0.146	2.266	<0.05
Conifer 100 m	-0.183	0.124	-1.469	0.142
Graminoid 30 m	0.304	0.123	2.479	<0.05
Line density 2000 m	-0.534	0.251	-2.131	<0.05

Table 26. Estimated regression parameters, standard errors, z-values and P-values for the 2018 early winter binomial GLM for the Clear Creek caribou herd in Yukon, Canada.

	Estimate	Std. error	z value	P-value
Intercept	-2.423	0.082	-29.428	<0.05
Elevation	9.329	0.646	14.452	<0.05
Elevation^2	-8.483	0.596	-14.250	<0.05
Disturbance 1000 m	-0.027	0.081	-0.330	0.74
Slope	0.494	0.150	3.297	<0.05
Slope^2	-0.713	0.159	-4.490	<0.05
Deciduous/mixed 250 m	-2.276	0.459	-4.962	<0.05
Shrublands 250 m	-0.479	0.130	-3.692	<0.05
Grasslands 250 m	0.333	0.096	3.472	<0.05
Wetland 250 m	0.817	0.101	8.067	<0.05
Non-vegetated 250 m	-0.180	0.201	-0.897	0.370
Lichen 30 m	0.407	0.052	7.819	<0.05
Conifer 250 m	0.468	0.048	9.642	<0.05
Graminoid 250 m	0.235	0.040	5.878	<0.05
Elevation : Disturbance 1000 m	2.006	0.753	2.663	<0.05
Elevation^2 : Disturbance 1000 m	-1.479	0.682	-2.168	<0.05

Table 27. Estimated regression parameters, standard errors, z-values and P-values for the 2018 late winter binomial GLM for the Clear Creek caribou herd in Yukon, Canada.

	Estimate	Std. error	z value	P-value
Intercept	-4.509	0.337	-13.389	<0.05
Elevation	30.023	2.963	10.133	<0.05
Elevation^2	-29.312	2.845	-10.304	<0.05
Line density 3000 m	-3.103	0.714	-4.347	<0.05
Slope	0.634	0.195	3.258	<0.05
Slope^2	-0.709	0.205	-3.457	<0.05
Deciduous/mixed 100 m	-2.616	0.590	-4.433	<0.05
Shrublands 100 m	-0.839	0.169	-4.968	<0.05
Grasslands 100 m	0.191	0.118	1.612	0.107
Wetland 100 m	0.940	0.119	7.917	<0.05
Non-vegetated 100 m	-1.775	0.372	-4.777	<0.05
Lichen 30 m	0.372	0.064	5.809	<0.05
Burns ≤50 yrs	-0.093	0.048	-1.950	0.05
Elevation : Line density 3000 m	24.639	6.279	3.924	<0.05
Elevation^2 : Line density 3000 m	-22.366	5.988	-3.735	<0.05

Table 28. Estimated regression parameters, standard errors, z-values and P-values for the 2018 summer binomial GLM for the Clear Creek caribou herd in Yukon, Canada.

	Estimate	Std. error	z value	P-value
Intercept	-5.746	0.399	-14.405	<0.05
Elevation	15.786	2.108	7.490	<0.05
Elevation^2	-11.055	1.647	-6.714	<0.05
RoadsTrails 1000 m	-2.697	0.804	-3.357	<0.05
Slope	0.718	0.197	3.639	<0.05
Slope^2	-0.690	0.171	-4.037	<0.05
Deciduous/mixed 100 m	-1.315	0.552	2.380	<0.05
Shrublands 100 m	0.017	0.155	0.109	0.913
Grasslands 100 m	0.344	0.151	2.274	<0.05
Wetland 100 m	0.310	0.179	1.732	0.083
Non-vegetated 100 m	0.284	0.163	1.737	0.082
Lichen 30 m	0.513	0.087	5.871	<0.05
Conifer 100 m	-1.185	0.059	-20.107	<0.05
Graminoid 30 m	0.159	0.057	2.770	<0.05
Elevation : RoadsTrails 1000 m	14.189	4.188	3.388	<0.05
Elevation^2 : RoadsTrails 1000 m	-10.829	3.257	-3.325	<0.05

Table 29. Estimated regression parameters, standard errors, z-values and P-values for the 2018 fall rut binomial GLM for the Clear Creek caribou herd in Yukon, Canada.

	Estimate	Std. error	z value	P-value
Intercept	-5.061	6.831	-0.741	0.459
Elevation	11.722	1.806	6.489	<0.05
Elevation^2	-7.769	1.459	-5.324	<0.05
Slope	0.530	0.284	1.865	0.062
Slope^2	-0.931	0.267	-3.489	<0.05
Deciduous/mixed 100 m	-1.522	1.053	-1.445	0.148
Shrublands 100 m	0.545	0.224	2.434	<0.05
Grasslands 100 m	0.576	0.215	2.685	<0.05
Wetland 100 m	0.835	0.247	3.378	<0.05
Non-vegetated 100 m	1.126	0.272	4.133	<0.05
Deciduous Shrub 100 m	-0.185	0.080	-2.300	<0.05
Graminoid 100 m	0.904	0.105	8.607	<0.05
Mining 500 m	-1.766	53.877	-0.033	0.974